Ramavath Sriram Naik

cs24btech11051@iith.ac.in | ramavathsriram06@gmail.com | +91 9494626880 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.86
Narayana Junior College, Adibatla	Class XII (Telangana Board of Intermediate Education)	2024	988.00%
Krishnaveni Talent School , Devarakonda	Class X (Telangana State Board of Secondary Education)	2022	9.80%

PROJECTS

Arbitrary Precision Arithmetic Library

- Developed a Java library to perform arbitrary-precision arithmetic on integers and floating-point numbers using custom string-based algorithms, eliminating limitations of primitive data types.
- Implemented core arithmetic operations including addition, subtraction, multiplication, division, and modulus without relying on Java's built-in big-number libraries.
- Designed modular AInteger and AFloat classes following Object-Oriented Programming principles, enabling reusable and maintainable arithmetic operations.
- Implemented sign handling, decimal alignment, carry/borrow propagation, and precision management to ensure correctness across large numeric computations.
- Validated arithmetic correctness through comprehensive unit testing across edge cases, including large operands, negative values, and high-precision decimal inputs.

Enhanced xv6 Operating System Kernel (RISC-V)

- Extended the xv6 operating system kernel by designing and implementing custom system calls for process management, parent-child relationships, kernel statistics, and system call accounting.
- Designed and implemented a System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ) scheduler, supporting four priority queues, dynamic priority adjustment, starvation prevention through global priority boosting, and runtime scheduling statistics.
- Developed a virtual memory subsystem featuring Clock-based page replacement, lazy allocation support, frame table management, swap space handling, and scheduler-aware page eviction to improve memory utilization under constrained physical memory.
- Implemented disk-backed swapping by replacing in-memory swap with persistent storage, integrating configurable FCFS and SSTF disk scheduling algorithms with simulated disk latency and request queue management.
- Simulated RAID 0, RAID 1, and RAID 5 storage mechanisms for swap operations, including striping, mirroring, parity generation, and fault-tolerant data reconstruction.
- Built comprehensive kernel instrumentation and user-space test programs to validate scheduling, virtual memory, disk I/O, and process statistics, ensuring stable execution without kernel panics.

5-Stage Pipelined RISC-V Simulator (C++, Qt)

- Designed and implemented a 5-stage pipelined RISC-V CPU simulator supporting Instruction Fetch, Decode, Execute, Memory, and Write-Back stages with cycle-accurate execution.
- Implemented hazard detection, data forwarding, pipeline stalls, bubble insertion, and branch flushing to correctly resolve data and control hazards in a pipelined processor.
- Developed multiple execution modes, including single-cycle, pipelined, forwarding-enabled, and static/dynamic (1-bit) branch prediction, enabling comparison of pipeline performance and correctness.
- Built a Qt-based interactive GUI featuring an assembly editor, register and memory viewers, execution statistics, step/run debugging, and real-time visualization of processor state.
- Implemented instruction decoding, immediate generation, ALU control, and memory access logic supporting the RV32I instruction set, enabling execution of complete assembly programs with detailed performance metrics.

SKILLS

- **Programming Languages** : C, C++, Java, Python, Html, Css, Java Script, SQL
- **FrameWork / Libraries** : React, Next.js
- **Tools** : Git/Github, SolidEdge

RELEVANT COURSES

- **Mathematics** : Linear Algebra, Discrete Maths for Computer Science, Calculus - I & II, Probability & Random Variables, Introduction to Statistics, Transform Techniques
- **Computer Science** : Algorithms, Operating Systems - I & II, Data Structures, Computer Architecture, Database Management Systems - I & II, Theory of Computations, Software Development Fundamentals, Programming in C, Compiler - I

POSITIONS OF RESPONSIBILITY

Teaching Assistant (TA)

- Working as a teaching assistant for the course Computer Architecture

Web Coordinator - Milan

- Managed the official event website, coordinated with organizing teams, and ensured timely updates and smooth online operations.

Web Coordinator - FCC

- Maintained the club website, published event content, and provided technical support for online activities.